RISC-V Stage 1

Group 1: RISC-y Designers

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EE 652 : IC Design Laboratory

Team Members

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Ibex Flow

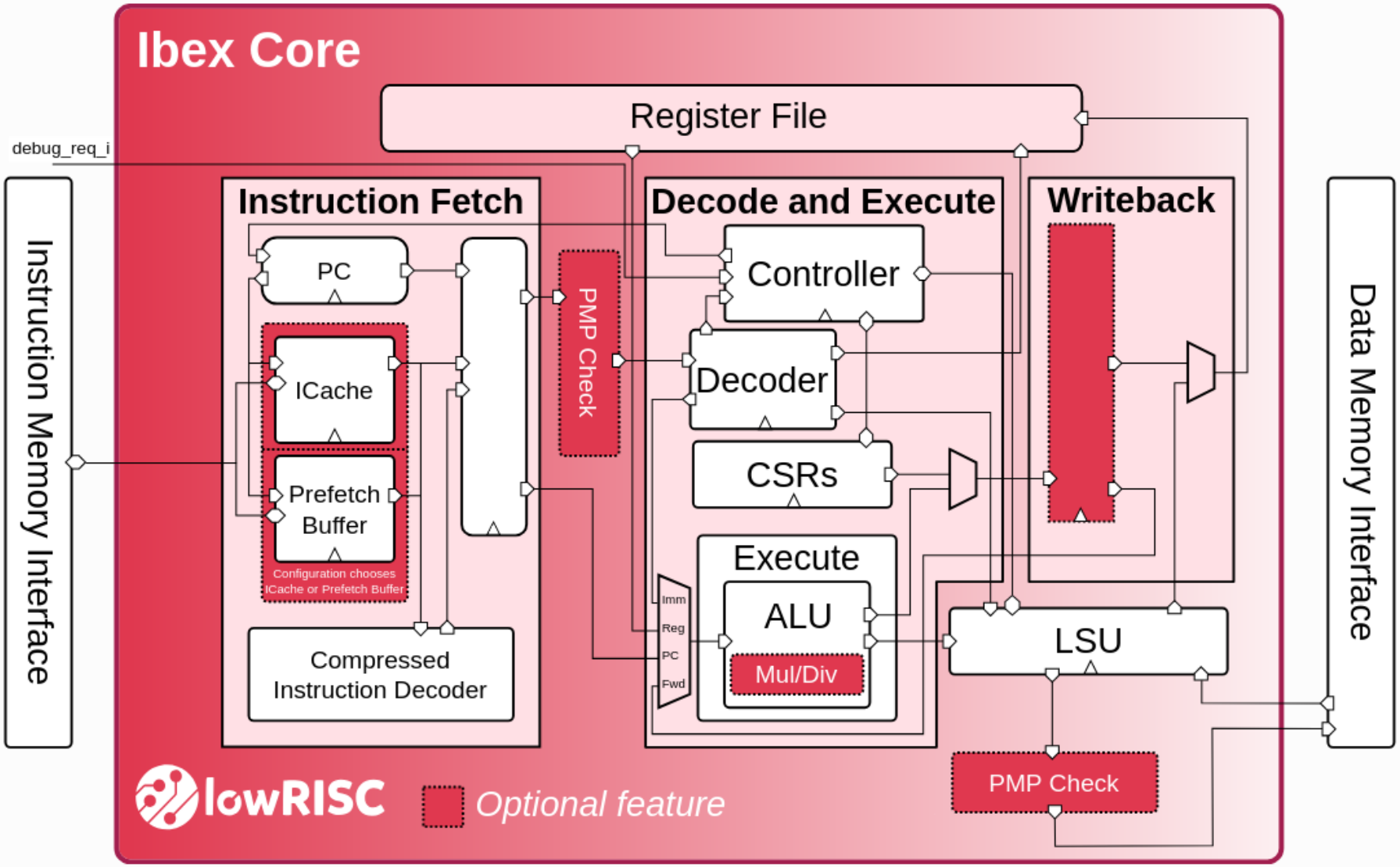
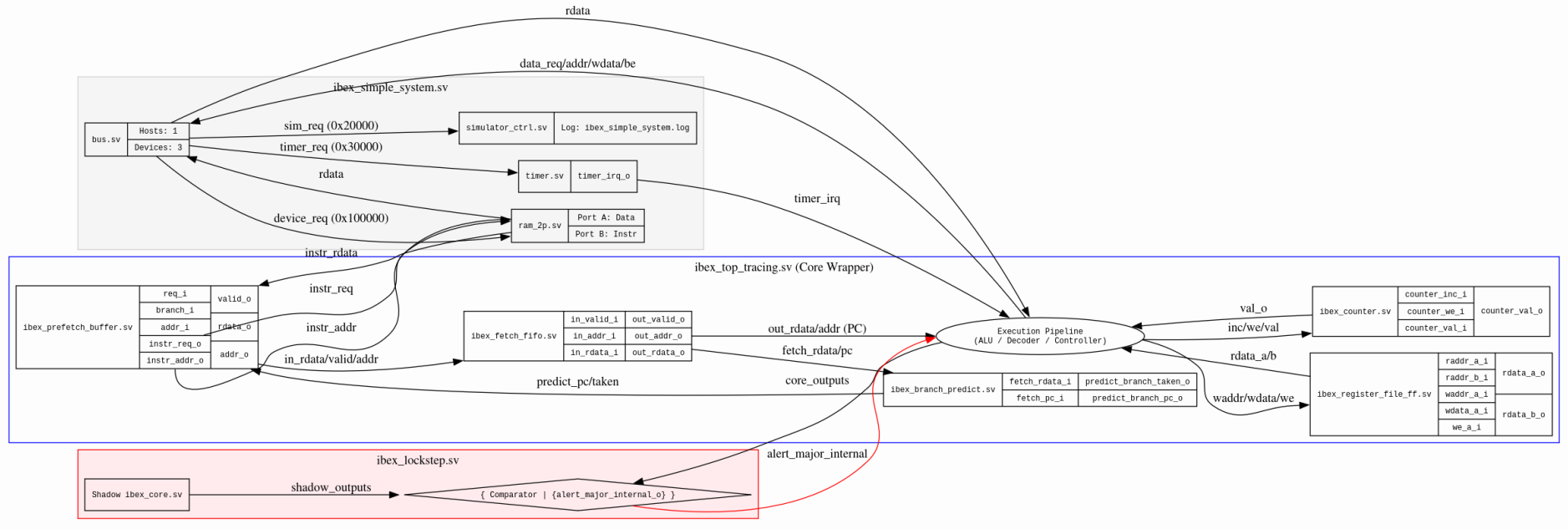


Figure 1 *Ibex Pipeline*

Ibex Flow


rdata
data_req/addr/wdata/be
ibex_simple_system.sv
bus.sv
Hosts: 1
Devices: 3
sim_req (0x20000)
simulator_ctrl.sv
Log: ibex_simple_system.log
timer_req (0x30000)
timer.sv
timer_irq_o
rdata
timer_irq
device_req (0x100000)
ram_2p.sv
Port A: Data
Port B: Instr
instr_rdata
ibex_top_tracing.sv (Core Wrapper)
instr_req
ibex_prefetch_buffer.sv
req_i
branch_i
addr_i
instr_req_o
instr_addr_o
valid_o
rdata_o
addr_o
instr_addr
in_rdata/valid/addr
ibex_fetch_fifo.sv
in_valid_i
out_valid_o
in_addr_i
out_addr_o
in_rdata_i
out_rdata_o
out_rdata/addr (PC)
fetch_rdata/pc
Execution Pipeline
(ALU / Decoder / Controller)
predict_pc/taken
core_outputs
ibex_branch_predict.sv
fetch_rdata_i
predict_branch_taken_o
fetch_pc_i
predict_branch_pc_o
val_o
inc/we/val
ibex_counter.sv
counter_inc_i
counter_we_i
counter_val_i
counter_val_o
rdata_a/b
waddr/wdata/we
ibex_register_file_ff.sv
raddr_a_i
raddr_b_i
waddr_a_i
wdata_a_i
we_a_i
rdata_a_o
rdata_b_o
alert_major_internal
ibex_lockstep.sv
Shadow ibex_core.sv
shadow_outputs
{ Comparator | {alert_major_internal_o} }

Work Distribution (1/2)

- **Abhinav Goud:** ibex_lockstep, ibex_prefetch_buffer, ibex_register_file_ff, ibex_counter, ibex_branch_predict, ibex_core.f, ibex_simple_system, ibex_fetch_fifo (1904 lines).
- **Akshat Barnwal:** ibex_core, ibex_register_file_latch, ibex_csr (2246 lines).
- **Nishi Shah:** ibex_cs_registers, ibex_top_tracing, ibex_dummy_instr (2198 lines).
- **Aditya Patil:** ibex_alu, ibex_multdiv_slow, ibex_register_file_fpga (1873 lines).
- **Ananya Balike:** ibex_top, ibex_multdiv_fast, ibex_wb_stage (2197 lines).

Work Distribution (2/2)

- **Hasan Ali:** ibex_decoder, ibex_load_store_unit (1836 lines).
- **Pavitr Chandwani:** ibex_tracer, ibex_tracer_pkg, ibex_fetch_fifo (1832 lines).
- **Chinmay Pendse:** ibex_icache, ibex_pkg, ibex_pmp, Instruction-wise analysis (2193 lines).
- **Aditya Prachand:** ibex_id_stage, ibex_if_stage, ibex_ex_block (2210 lines).
- **Vivek Raj:** ibex_controller, ibex_compressed_decoder (1791 lines).

Summary of Modules

Total Modules Analyzed: 32 files (31 SystemVerilog modules/packages and 1 file list)

- **Core Integration:** `ibex_top.sv` and `ibex_core.sv` wrap the structural backbone.
- **Pipeline Stages:**
  - IF: Instruction fetch and PC selection.
  - ID: Decodes instructions and manages register file ops.
  - EX: The computational engine enclosing the ALU.
  - LSU: Memory engine managing loads/stores.
  - WB: Optional Writeback stage boundary.
- **Execution Units & Memory:** `ibex_alu.sv`, `ibex_pmp.sv`, and Register Files (FF, Latch, FPGA).

Current Architecture: RV32I Pipelining Status

- **Pipelined Segments:**
  - The Instruction Decoder (`ibex_decoder.sv`) is pipelined and continuously pushes to the queue.
- **Unpipelined Segments (Bottlenecks):**
  - **Load/Store Unit (LSU):** Uses a multi-state FSM (`WAIT_GNT`, etc.) and asserts a `busy_o` signal to stall the entire pipeline during memory access.
  - **ID/EX State Machine:** Shifts to a `MULTI_CYCLE` state to explicitly stall the pipeline for loads.
  - **Writeback (WB) Stage:** Defaults to a 2-stage mode (`WritebackStage = 0`), acting as a combinational pass-through with no pipeline registers.

Tentative Plan: Achieving a Fully Pipelined RV32I

- **Register the WB Stage:** Enable the `WritebackStage = 1` parameter to instantiate flip-flops, converting the core from a 2-stage to a true 3-stage pipeline.
- **Redesign the LSU:** Modify the Load/Store Unit's FSM so it can issue requests continuously without asserting the `busy_o` stall signal for every transaction.
- **Upgrade Hazard Logic:** Overhaul the ID stage's `MULTI_CYCLE` state machine.
- Instead of relying on stalls for multi-cycle instructions, implement robust data forwarding paths to resolve Read-After-Write (RAW) hazards dynamically.

Future Scope: Pipelining the RV32IM Extension

- **The 'M' Extension Challenge:** If we expand to RV32IM, the Multiplier and Divider units (`ibex_multdiv_slow.sv` / `fast.sv`) introduce major unpipelined blocks.
- **Iterative Stalls:** Currently, division uses an iterative restoring algorithm taking up to 34 clock cycles, completely stalling the ID stage.
- **Action Required:** To pipeline RV32IM, we would need to unroll these iterative FSMs into a fully pipelined multiplier/divider array that can accept new operands on every clock cycle without asserting `mult_hold` or `div_hold`.

The Mammoth Task: Getting Skeleton Ibex Working

- **Massive Codebase Integration:** Analyzing and structurally wiring complex SystemVerilog code split across 30+ separate modules to implement an open-source 32-bit CPU.
- **Dependency Resolution:** Sourcing the repository from GitHub and fulfilling intricate Python dependencies just to generate the base SV codes.
- **Complex Parameterization:** The top-level wrapper (`ibex_top.sv`) alone contains over 40 parameters that must be correctly configured to get a functional baseline processor.
- **Intricate Interconnections:** Understanding the deep routing between the IF, ID, EX, LSU, and WB blocks, and manually ensuring that pipeline control signals behave correctly.